NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 0

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

Layer "X" in HART incorporates channel hopping and channel blacklisting to increase reliability and security. What is X?

a. Physical layer

b. Data Link layer

c. Network layer

d. Transport layer

Correct Answer: b. Data Link layer

Detailed Solution: HART Data link layer, derived from the IEEE 802.15.4 standard, incorporates channel hopping and channel blacklisting to increase reliability and security.

See lecture 11@ 07:25

QUESTION 2:

What does ISA stand for in ISA 100.11A?

a. International Society of Automation
b. Industrial Society of Automation
c. International Standards of Automation
d. Industrial Standards of Automation

Correct Answer: a. International Society of Automation

Detailed Solution: ISA100.11A is a wireless networking technology standard developed by the International Society of Automation (ISA).
See lecture 13@ 14:30

QUESTION 3:

"X" is a web transfer protocol based on Request-Response model designed for M2M applications. What is "X"?

a. AMQP
b. CoAP
c. Both AMQP and CoAP
d. None of these

Correct Answer: b. CoAP

Detailed Solution: CoAP – Constrained Application Protocol is a web transfer protocol. Ii is based on request-response model and designed for M2M applications. See lecture 6 @ 01:20

QUESTION 4:

In case of XMPP protocol, binary data must be first encoded to _____ before transmission.

a. base128

b. base32

c. base256

d. base64

Correct Answer: d. base64

Detailed Solution: Any XML character data contained within the XML elements used in XMPP protocol MUST be encoded using base64. See lecture 7 @ 16:50

QUESTION 5:

Which of the following is NOT a constraint for UAV network?

a. Prone to environmental effects

b. Changing topology

c. Frequent link breakages

d. None of these

Correct Answer: d. None of these

Detailed Solution: UAV network constraints include frequent link breakages, prone to environmental effects, prone to malfunction, and changing topology or node position.

See lecture 19 @ 09:43

QUESTION 6:

Is a star topology-based UAV network self-configuring?

a. Yes
b. No

Correct Answer: b. No

Detailed Solution: A star topology-based UAV network is not self-configuring.

See lecture 19 @ 13:05

QUESTION 7:

Which of the following can be used as a position sensor?

a. RFM3200
b. Potentiometer
c. DHT11
d. PS2591

Correct Answer: b. Potentiometer

Detailed Solution: Potentiometer works by varying the position of a sliding contact across a uniform resistance. It is often used as a position sensor. See lecture 3 (Sensing)

QUESTION 8:

Which of the following is not an inbuilt function of servo library?

a. Sweep()
b. Knob()
c. Move()
d. None of the above

Correct Answer: c. Move()

Detailed Solution: Knob() and Sweep() are the inbuilt functions.
See lecture 25 @ 11:15

QUESTION 9:

Which of the following is true for "= =" operator in python programming?

a. compares the values of both the operands and checks for value equality
b. checks whether both the operands refer to the same object
c. checks if the value is found in the sequence
d. None of the above

Correct Answer: a. compares the values of both the operands and checks for value equality

Detailed Solution: As per basics of Python programming.

QUESTION 10:

Which of the following is not a component of a switch?

a. Application
b. Operating System
c. Specialized hardware
d. None of these

Correct Answer: d. None of these

Detailed Solution: Since switches in non-SDN do not have global view of the network, they use a decentralized method for forwarding the data. (Please refer Lecture 33@9:13)

QUESTION 11:

Which of the following is not a function of the Application Plane in Software-Defined Network (SDN) architecture?

a. Routing
b. Security
c. Traffic Engineering
d. Forwarding

Correct Answer: d. Forwarding

Detailed Solution: The Data Plane in SDN is responsible for data forwarding. (Please refer Lecture 33@12:25)

QUESTION 12:

Who among the following are benefited on using smart grids?

a. Consumers

b. Stakeholders

c. Both consumers and stakeholders

d. Neither consumers nor stakeholders

Correct Answer: c. Both consumers and stakeholders

Detailed Answer: Both consumers and stakeholders are benefited on using smart grids. (Please refer Lecture 51@13:54)

QUESTION 13:

Micro-grids can detach from the main smart grids at any point of time without disrupting the core grid.

a. True
b. False

Correct Answer: a. True

Detailed Answer: Micro-grids can detach from the main smart grids at any point of time without disrupting the core grid.

QUESTION 14:

Proactive rule placement in Mobi-Flow depends on which of the following user characteristic?

a. User portfolio
b. User movement
c. User handoff
d. User type

Correct Answer: b. User movement

Detailed Answer: The proactive rule placement depends on the user movement and position in the future course of time. (Please refer Lecture 36@15:18)

QUESTION 15:

Fill in the blank: Typically, cloud computing have _____ components?

a. 4
b. 5
c. 6
d. 7

Correct Answer: c. 6

Detailed Answer: Cloud computing has 6 components. They are clients, services, applications, platforms, storage, and infrastructure. (Please refer Lecture 37@23:31)

**********END**********

Introduction to

Internet of Things

Assignment-Week 1

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

In the context of IoT, what is the meaning of ubiquitous?

a. Spreading widely throughout
b. Present everywhere
c. Countless
d. Uncontrollable

Correct Answer: b. Present everywhere

Detailed Solution: Ubiquitous refers to presence everywhere or in many places especially simultaneously. Another word for it is omnipresence. (Please refer Lecture 1@9:41)

QUESTION 2:

Typical IoT networks has to support the existence of intermittent connectivity. Which one of the following is a characteristic of intermittent connectivity?

a. Controlled network drops
b. No network drops
c. Random/uncontrolled network drops
d. Continuous connection

Correct Answer: c. Random/uncontrolled network drops

Detailed Solution: Intermittent connectivity refers to random loss of connections due to mobility, environment factors, and others. (Please refer Lecture 1@20:21)

QUESTION 3:

In a typical IoT deployment, which of the following is true with respect to cardinality of each type mentioned below?

a. Applications > Smart Systems > Sensors
b. Sensors > Applications > Smart Systems
c. Smart Systems > Sensors > Applications
d. Sensors > Smart Systems > Applications

Correct Answer: d. Sensors > Smart Systems > Applications

Detailed Solution: It is expected that there will be trillions of sensors, billions of smart devices, and millions of applications. (Please refer Lecture 1@30:21)

QUESTION 4:

In the IoT connectivity layers, the service providers are responsible for which of the following services?

a. Services
b. Local connectivity
c. Global connectivity
d. None of the above

Correct Answer: b. Local connectivity

Detailed Solution: The service providers are responsible for the local connectivity. (Please refer Lecture 1@32:11)

QUESTION 5:

Which of the following uses point-to-point communication?

a. Internet of Things (IoT)
b. Machine-to-Machine (M2M) communications
c. Cyber-Physical Systems (CPS)
d. Web-of-Things (WoT)

Correct Answer: b. Machine-to-Machine (M2M) communications

Detailed Solution: M2M is peer-to-peer/point-to-point communications. (Please refer Lecture 1@32:26)

QUESTION 6:

Which of the following enables access and control over IoT resources using web technologies?

a. Machine-to-Machine (M2M) communications
b. Cyber-Physical Systems (CPS)
c. Web-of-Things (WoT)
d. None of these

Correct Answer: c. Web-of-Things (WoT)

Detailed Solution: WoT allows access to IoT through web technologies for the users. (Please refer Lecture 1@34:26)

QUESTION 7:

Which of the following terminologies is most suitable for cyber-physical systems in context of IoT?

a. Scale
b. Autonomy
c. Environment
d. Synthesis

Correct Answer: b. Autonomy

Detailed Solution: Cyber-physical systems need autonomy for making smart decisions and executing collaborative tasks. (Please refer Lecture 1@36:27)

QUESTION 8:

Fill in the blank:

The _____ is responsible for connecting the IoT LAN to a WAN and then to the Internet.

a. IoT LAN
b. IoT Proxy
c. IoT Node
d. IoT Gateway

Correct Answer: d. IoT Gateway

Detailed Solution: IoT Gateway connects the IoT LAN to a WAN to the Internet. It can implement several LANs and WANs. (Please refer Lecture 2@3:11)

QUESTION 9:

Under Unique Local Addresses (ULA), what happens in case of unavailability of the pre-assigned local address when a device migrates from one LAN to another?

a. Use same local address but different MAC address
b. Do not allow networking to that device
c. Assign new available local address
d. Disconnect previously existing device

Correct Answer: c. Assign new available local address

Detailed Solution: In case of unavailability, the device will be assigned a new local address and the gateway will be updated.

QUESTION 10:

True or False?

Changes in network prefixes does not affect the remote anchor point to which the IoT gateways are attached.

a. True
b. False

Correct Answer: a. True

Detailed Solution: Remote anchor points are immune to changes in network prefixes. (Please refer Lecture 2@11:01)

QUESTION 11:

What is a device called that detects or measures a physical property?

a. Sensors
b. Actuators
c. Nodes
d. Users

Correct Answer: a. Sensors

Detailed Solution: Sensors detect or measure changes in physical properties and environments. (Please refer Lecture 3@3:25)

QUESTION 12:

Which of the following may be referred to as a transducer?

a. Sensor
b. Actuator
c. Neither sensor nor actuator
d. Both sensor and actuator

Correct Answer: d. Both sensor and actuator

Detailed Solution: Transducers cover both sensors and actuators. (Please refer Lecture 3@11:15)

QUESTION 13:

Which of the following types of sensors may be identified when categorized based on output?

a. Analog
b. Scalar
c. Vector
d. Multimedia

Correct Answer: a. Analog

Detailed Solution: Based on output, sensors may be classified as analog and digital. (Please refer Lecture 3@13:00)

QUESTION 14:

Which IoT component is responsible for hosting analytics on the data?

a. Things
b. Local network
c. Backend Services
d. Internet

Correct Answer: c. Backend services

Detailed Solution: The backend services are responsible for providing/performing analytics on the data. (Please refer Lecture 5@7:21)

QUESTION 15:

Typically, how many categories (application-specific) is IoT classified into?

a. 1
b. 2
c. 3
d. 4

Correct Answer: b. 2

Detailed Solution: IoT may be classified into industrial and consumer IoT. (Please refer Lecture 5@17:51)

**********END**********

Introduction to

Internet of Things

Assignment-Week 2

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

With respect to protocol design, which of the following category best defines WiFi, Bluetooth, and LPWAN?

a. Connectivity
b. Discovery
c. Transport
d. Multi-layer framework

Correct Answer: c. Transport

Detailed Solution: WiFi, Bluetooth, and LPAN falls under Communication or Transport protocol design. (Please refer Lecture 6@1:01)

QUESTION 2:

Which of the following refers to data protocol?

a. MQTT
b. EPC
c. RPL
d. TR-069

Correct Answer: a. MQTT

Detailed Solution: MQTT, CoAP, AMQP, Websocket, and others are referred to as data protocols. (Please refer Lecture 1@1:01)

QUESTION 3:

In MQTT, who controls the publish-subscribe messaging pattern?

a. Publisher
b. Subscriber
c. Broker
d. Service provider

Correct Answer: c. Broker

Detailed Solution: In MQTT, the broker controls the publish-subscribe messaging pattern. (Please refer Lecture 6@3:31)

QUESTION 4:

In what format are the topics in MQTT updated by the broker?

a. Packet
b. Message
c. Bit
d. Signal

Correct Answer: b. Message

Detailed Solution: The topics are updated and distributed in the form of messages by the broker. (Please refer Lecture 6@3:31)

QUESTION 5:

Which wildcard would you use for getting messages from an entire subtree of topics in MQTT?

a. +
b. -
c. *
d. #

Correct Answer: d. #

Detailed Solution: The wildcard + is used for one level hierarchy and # is used for the entire subtree. (Please refer Lecture 6@12:11)

QUESTION 6:

How many messaging modes does CoAP have?

a. 1
b. 2
c. 3
d. 4

Correct Answer: d. 4

Detailed Solution: Confirmable, Non-confirmable, piggyback, and separate are the 4 messaging modes in CoAP. (Please refer Lecture 7@6:51)

QUESTION 7:

What message request does CoAP use for updates?

a. GET
b. PUT
c. PUSH
d. DELETE

Correct Answer: c. PUSH

Detailed Solution: CoAP uses similar messages as HTTP. (Please refer Lecture 7@7:51)

QUESTION 8:

Which of the following is true for piggyback messages in CoAP?

a. ACK
b. ACK+DATA
c. CON+DATA
d. ACK+CON

Correct Answer: b. ACK+DATA

Detailed Solution: IoT Gateway connects the IoT LAN to a WAN to the Internet. It can implement several LANs and WANs. (Please refer Lecture 2@3:11)

QUESTION 9:

True or False?

XMPP is suitable for cloud computing deployments and does not support M2M or P2P communications.

a. True
b. False

Correct Answer: b. False

Detailed Solution: XMPP is suitable for cloud computing deployments and supports M2M or P2P communications across a diverse set of networks. (Please refer Lecture 7@12:12)

QUESTION 10:

Which of the following is not true for XMPP?

a. Centralized
b. No royalties necessary
c. Support for security
d. Support for interoperability

Correct Answer: a. Centralized

Detailed Solution: XMPP is decentralized and does not need central servers. (Please refer Lecture 7@13:47)

QUESTION 11:

What is the basic unit of data in the AMQP protocol?

a. Packet
b. Bit
c. Frame
d. Segment

Correct Answer: c. Frame

Detailed Solution: The basic unit of data in AMQP is a frame. (Please refer Lecture 8@0:43)

QUESTION 12:

Which IEEE 802.15.4 variant is used for industrial applications?

a. D
b. E
c. F
d. G

Correct Answer: b. E

Detailed Solution: IEEE 802.15.4/E is used for industrial applications. (Please refer Lecture 9@9:11)

QUESTION 13:

What network topology is popularly used in Zigbee networks?

a. Bus
b. Star
c. Mesh
d. Ring

Correct Answer: c. Mesh

Detailed Solution: Zigbee are popularly used in the form of mesh wireless networks. (Please refer Lecture 9@16:16)

QUESTION 14:

Which of the following is the root in a Zigbee network?

a. Zigbee coordinator
b. Zigbee router
c. Zigbee end device
d. Any of these

Correct Answer: a. Zigbee coordinator

Detailed Solution: The Zigbee coordinator acts as the root of a Zigbee network and initiates it. (Please refer Lecture 9@22:51)

QUESTION 15:

In the LOADng routing protocol, who responds to the RREQ messages?

a. Destination
b. Relays
c. Sensor
d. Both destination and relays

Correct Answer: a. Destination

Detailed Solution: Only destination responds to RREQ messages in LOADng protocol. (Please refer Lecture 10@12:53)

**********END**********

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of ThingsAssignment-Week 3

TYPE OF QUESTION:MCQ/MSQ

Number ofquestions:15 **Total marks: 15 X 1= 15**

QUESTION 1:

Layer "X" incorporates channel hopping and channel blacklisting to increase reliability and security. What is "X"?

a. HART Physical layer
b. HART Data Link layer
c. HART Network layer
d. HART Transport layer

Correct Answer: b. HART Data Link layer

Detailed Solution:HART Data link layer, derived from the IEEE 802.15.4 standard, incorporates channel hopping and channel blacklisting to increase reliability and security.
See lecture 11@ 07:25

QUESTION 2:

The transmission in HART is synchronized using ____ slots.

a. 100 microseconds
b. 100 milliseconds
c. 10 milliseconds
d. 10 microseconds

Correct Answer: c. 10 milliseconds

Detailed Solution: In order to control congestion in HART, transmissions are synchronized using 10 milliseconds slots.
See lecture 11@ 11:50

QUESTION 3:

For control applications, which of the following ISA 100.11A usage classes are defined?

a. Class 0
b. Class 1&2
c. Class 1,2&3

d. Class 4&5

Correct Answer: c. Class 1,2&3

Detailed Solution: Usage classes 1,2&3 are defined for control applications. 1: closed loop regulatory control; 2: closed loop supervisory control; 3: open loop control.
See lecture 13@ 19:20

QUESTION 4:

The deterministic communication in Data Link Layer of HART is achieved by _____

a. Channel hopping
b. Channel blacklisting
c. Super-frames
d. Modulation

Correct Answer: c. Super-frames

Detailed Solution: Collison-free and deterministic communication in HART Data Link Layer achieved by super-frames and TDMA.
See lecture 11@ 6:30

QUESTION 5:

What does ISA stand for in ISA 100.11A?

a. International Society of Automation
b. Industrial Society of Automation
c. International Standards of Automation
d. Industrial Standards of Automation

Correct Answer: a. International Society of Automation

Detailed Solution:ISA100.11A is a wireless networking technology standard developed by the International Society of Automation (ISA).

See lecture 13@ 14:30

QUESTION 6:

"Hop selection" is supported by _____ in Bluetooth technology.

a. Baseband layer

b. L2CAP

c. Both (a) and (b)

d. None of the above

Answer: a. Baseband layer

Detailed Solution: The baseband layer of Bluetooth protocol stack supports services like error correction, data whitening, hop selection, and security.
See lecture 12@11:15

QUESTION 7:

Which of the following is NOT a scheme for re-establishing the connectivity between dumb nodes with other nodes in a wireless sensor networks?

a. CoRD
b. CaRD
c. CoRAD
d. None of the above

Correct Answer: b. CaRD

Detailed Solution: CoRD and CoRAD are the schemesfor re-establishing the connectivity between dumb nodes with other nodes in a wireless sensor networks.
See lecture 15@ 10:50

QUESTION 8:

Which of the following technology does not use the standard 2.4 GHz ISM band?

a. ZigBee
b. Bluetooth
c. NFC
d. 6LowPAN

Correct Answer: c. NFC

Detailed Solution: NFC utilizes 13.56 MHz frequency of the ISM band.

QUESTION 9:

WirelessHART Network Manager handles _____-based network security.

a. Code
b. Collision
c. Time
d. Access

Correct Answer: a. code

Detailed Solution: WirelessHARTnetwork manager handles code-based network security and prevents unauthorized nodes from joining the network.
See lecture 11@ 13:08

QUESTION 10:

How many power-saving modes are there in Bluetooth technology?

a. One
b. Two
c. Three
d. Four

Correct Answer: c. Three

Detailed Solution: There are three power-saving modes in Bluetooth technology, namely, sniff, hold, and park.
See lecture 12@ 07:35

QUESTION 11:

Which layer provides protocol multiplexing capability?

a. Physical layer
b. Data link layer
c. Middleware layer
d. Application layer

Correct Answer: b. Data link layer

Detailed Solution: Logical link control and adaptation protocol (L2CAP) provides protocol multiplexing capability, which resides in the Data Link Layer.
See lecture 12@ 12:20

QUESTION 12:

What does RF4CE stand for in "ZigBee RF4CE"?

a. Radio Frequency for Consumer Electronics
b. Radio Frequency 4.0 Consumer Electronics
c. Radio Frequency for Controlled Environment
d. Radio Frequency 4.0 Controlled Environment

Correct Answer: a. Radio Frequency for Consumer Electronics

Detailed Solution: ZigBee RF4CE (Radio Frequency for Consumer Electronics) is a subset of ZigBee 3.0, developed to replace the infrared remote controls for consumer electronics (TVs, stereos) with radio-based controls.
See lecture 11@ 16:00

QUESTION 13:

NFC tags found in supermarket products are examples of _____ NFC.

a. Active

b. Passive

c. Both (a) and (b)

d. None of the above

Correct Answer: b. Passive

Detailed Solution: NFC tags found in supermarket products are passive devices. The smartphones are examples of active devices.
See lecture 11@ 16:40

QUESTION 14:

Which of the following utilizes Manchester channel encoding?

a. ZigBee
b. NFC
c. Zwave
d. None of the above

Correct Answer: c. Zwave

Detailed Solution: Zwave utilizes GFSK modulation and Manchester channel encoding.
See lecture 13@ 06:50

QUESTION 15:

What are the constraints on sensor nodes?

a. Dispensable
b. Autonomous
c. Low power consumption
d. All of these

Correct Answer: d. All of these

Detailed Solution: Any sensor node must be of small size, consuming low power, and be dispensable, autonomous, and adaptive to the environment.
See lecture 14@ 15:30

**********END**********

Introduction to

Internet of Things

Assignment-Week 4

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

Choose the appropriate answer about the parameter that the sensor detects:

a. Passive Infrared: Detect object, Ultrasonic: Distance
b. Passive Infrared: Distance, Ultrasonic: Detect object
c. Passive Infrared: Color, Ultrasonic: Detect sound
d. Passive Infrared: Detect sound, Ultrasonic: Color

Correct Answer: a. Passive Infrared: Detect object, Ultrasound: Distance

Detailed Solution: Passive Infrared: Detect object, Ultrasound: Distance. (Please refer Lecture 16@13:21)

QUESTION 2:

In typical wireless multimedia sensor network deployments, what is the relation between the number of camera (C) and scalar sensors (S)?

a. S < C
b. S > C
c. S = C
d. None of the above

Correct Answer: b. S > C

Detailed Solution: In typical wireless multimedia sensor network deployments, the number of camera (C) sensors is less than that of scalar sensors (S) (Please refer Lecture 16@17:21)

QUESTION 3:

Which of the following represents nano scale?

a. 10^{12}
b. 10^{-12}
c. 10^{-9}
d. 10^{9}

Correct Answer: c. 10^{-9}

Detailed Solution: 10^{-9} represents nano and 10^{-12} represents pico.

QUESTION 4:

In molecular communications, what works as a mediator between cells and vesicles?

a. Dendrite
b. Axon
c. Nucleus
d. Gap junction

Correct Answer: d. Gap junction

Detailed Solution: In molecular communications, gap junction works as a mediator between cells and vesicles. (Please refer Lecture 16@28:06)

QUESTION 5:

Typically, what frequency range do the nano nodes in a nano network operate/communicate?

a. THz
b. GHz
c. MHz
d. Hz

Correct Answer: a. THz

Detailed Solution: The nano nodes in a nano network usually operate in the 0.1-10 THz band. (Please refer Lecture 16@28:41)

QUESTION 6:

Which of the following relations is necessary for establishing that coverage implies connectivity?

a. Transmission range > 2*sensing range
b. Transmission range >= 2*sensing range
c. Transmission range < 2*sensing range
d. Transmission range <= 2*sensing range

Correct Answer: b. Transmission range >= 2*sensing range

Detailed Solution: Transmission range >= 2*sensing range is necessary to establish that coverage implies connectivity. (Please refer Lecture 17@4:25)

QUESTION 7:

In sensor networks, how many types of reporting exists?

a. 1
b. 2
c. 3
d. 4

Correct Answer: b. 2

Detailed Solution: In sensor networks, there are 2 types of reporting: event driven and on demand. (Please refer Lecture 17@8:06)

QUESTION 8:

Fill in the blank:

Coverage algorithms are centralized _____ decentralized _____ localized.

a. and, and
b. and, or
c. or, and
d. or, or

Correct Answer: c. or, and

Detailed Solution: Coverage algorithms are centralized or decentralized and localized. (Please refer Lecture 17@10:07)

QUESTION 9:

True or False?

The main objective of coverage in networking is to increase the lifetime of the network with maximum number of sensor nodes in an area.

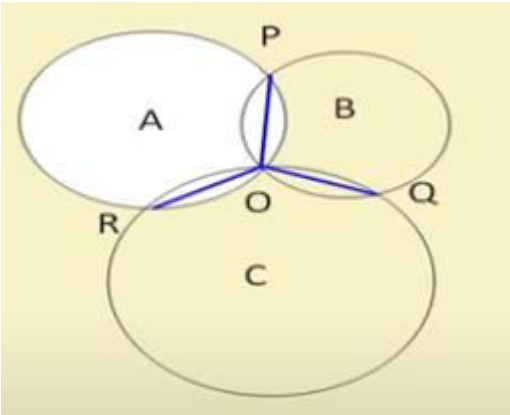
a. True
b. False

Correct Answer: b. False

Detailed Solution: The main objective of coverage in networking is to increase the lifetime of the network with minimum number of sensor nodes in an area. (Please refer Lecture 17@11:22)

QUESTION 10:

In the given diagram, if the nodes have the same sensing range (r_s), what is the distance between them in case of optimal coverage?



a. r_s
b. $3\ r_s$
c. $\sqrt{3}\ r_s$
d. $\sqrt{2}\ r_s$

Correct Answer: c. $\sqrt{3}\ r_s$

Detailed Solution: For equal sensing range, the distance between the sensor nodes on optimal placement for converage is $\sqrt{3}\ r_s$. (Please refer Lecture 17@20:23)

QUESTION 11:

True or False?

Sensor nodes are capable of collaborating with one another.

a. True
b. False

Correct Answer: a. True

Detailed Solution: Sensor nodes are capable of collaborating with one another. (Please refer Lecture 18@4:01)

QUESTION 12:

What device is responsible for collecting data from the sensor nodes and deliver it to sink nodes?

a. Sensor nodes
b. User devices
c. Data mules
d. Relays

Correct Answer: c. Data mules

Detailed Solution: Data mules are responsible for collecting data from the sensor nodes and delivering it to sink nodes. (Please refer Lecture 18@7:21)

QUESTION 13:

In UAV networks, what topology requires one node from each group to connect to the ground base station?

a. Star
b. Multi star
c. Hybrid star
d. Hierarchical star

Correct Answer: b. Multi star

Detailed Solution: The multi star network topology requires one node from each froup to connect to the base station. (Please refer Lecture 19@12:21)

QUESTION 14:

In UAV networks, what topology supports multi-point to multi-point communications?

a. Bus
b. Star
c. Ring
d. Mesh

Correct Answer: d. Mesh

Detailed Solution: In UAV networks, mesh topology supports multi-point to multi-point communications. (Please refer Lecture 19@13:51)

QUESTION 15:

True or False?

M2M is a peer to peer networking paradigm and does not support cross-platform communications.

a. True
b. False

Correct Answer: b. False

Detailed Solution: M2M was designed for cross-platform integration. (Please refer Lecture 20@4:21)

**********END**********

Introduction to

Internet of Things

Assignment-Week 5

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

What interoperability solution approach creates virtual maps among the physical devices of all middleware home networks?

a. Universal Middleware Hub
b. Universal Middleware Router
c. Universal Middleware Bridge
d. Universal Middleware Gateway

Correct Answer: c. Universal Middleware Bridge

Detailed Solution: Universal Middleware Bridge creates virtual maps among the physical devices of all middleware home networks. (Please refer Lecture 21@22:16)

QUESTION 2:

Which UMB interoperability component is responsible for converting physical devices into virtually abstracted ones?

a. UMB Adaptor
b. UMB Core
c. UMB Hypervisor
d. UMB Abstractor

Correct Answer: a. UMB Adaptor

Detailed Solution: UMB-A is responsible for converting physical devices into virtually abstracted ones. (Please refer Lecture 21@27:16)

QUESTION 3:

Which UMB interoperability component is responsible for routing the metadata messages to the destinations or other intermediate devices?

a. UMB Adaptor
b. UMB Core
c. UMB Hypervisor
d. UMB Router

Correct Answer: b. UMB Core

Detailed Solution: UMB Core routes the packets among the UMB-As or the destination. (Please refer Lecture 21@28:16)

QUESTION 4:

What does the routing component in UMB use for routing the metadata messages?

a. Flow Table
b. Routing Table
c. Middleware Routing Table
d. Middleware Flow Table

Correct Answer: c. Middleware Routing Table

Detailed Solution: UMB-C uses Middleware Routing Table for routing metadata messages among the devices. (Please refer Lecture 21@28:16)

QUESTION 5:

What type of input signals does an Arduino accept?

a. Analog
b. Digital
c. Neither analog nor digital
d. Both analog and digital

Correct Answer: d. Both analog and digital

Detailed Solution: Arduino accepts both analog and digital signals. (Please refer Lecture 22@5:11)

QUESTION 6:

What external hardware module is necessary for loading the programs into an Arduino board?

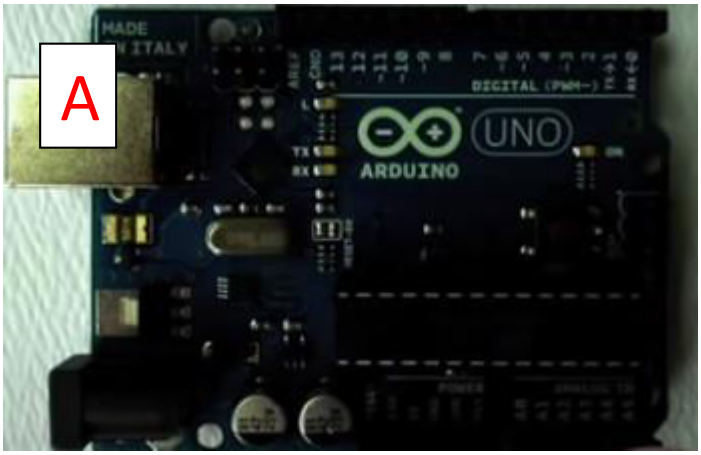
a. ESP8266
b. ESP32
c. ESP12e
d. None of these

Correct Answer: d. None of these

Detailed Solution: Arduino boards do not need any external hardware for loading programs into them. (Please refer Lecture 22@5:11)

QUESTION 7:

In the Arduino shown, what is A?



a. USB connector
b. Power connector
c. Ethernet/LAN connector
d. Analog reference

Correct Answer: a. USB connector

Detailed Solution: USB connector. (Please refer Lecture 22@7:41)

QUESTION 8:

What is a program written in Arduino called?

a. Stretch
b. Snippet
c. Service
d. Sketch

Correct Answer: d. Sketch

Detailed Solution: Program written in Arduino is called Sketch. (Please refer Lecture 22@12:21)

QUESTION 9:

Which function in Arduino is analogous to the main() function in C/C++?

a. setup()
b. loop()
c. __init__()
d. main()

Correct Answer: a. setup()

Detailed Solution: The setup() function in Arduino is analogous to main() function in C/C++. (Please refer Lecture 22@13:46)

QUESTION 10:

In Arduino, what function is used to find if a character is hexadecimal?

a. ishexdigit(Char)
b. isxdigit(Char)
c. isHexadecimalDigit(Char)
d. ishex(Char)

Correct Answer: c. isHexadecimalDigit(Char)

Detailed Solution: In Arduino, isHexadecimalDigit (Char) function is used to find is a character is hexadecimal. (Please refer Lecture 22@20:06)

QUESTION 11:

In Arduino, what function is used to manipulate time or induce delays?

a. sleep()
b. timer()
c. delay()
d. stop()

Correct Answer: c. delay()

Detailed Solution: The delay() function in Arduino manipulates time or induce delays. (Please refer Lecture 22@20:06)

QUESTION 12:

Choose the right option for if/conditional operator.

a. Val = (condition)?(Statement 1):(Statement 2)
b. Val = (condition)?(Statement 2):(Statement 1)
c. Val = (condition):(Statement 1)?(Statement 2)
d. Val = (condition):(Statement 2)?(Statement 1)

Correct Answer: a. Val = (condition)?(Statement 1):(Statement 2)

Detailed Solution: Conditional operator may also be written as Val = (condition)?(Statement 1):(Statement 2). (Please refer Lecture 23@2:01)

QUESTION 13:

What is a loop inside a loop called?

a. Loop
b. Infinite loop
c. Nested loop
d. Recursive loop

Correct Answer: c. Nested loop

Detailed Solution: A loop inside a loop is called Nested loop. (Please refer Lecture 23@2:51)

QUESTION 14:

What function is useful for generating the same set of random numbers every time a program is run?

a. randomgen()
b. randomint()
c. randomseed()
d. constrandom()

Correct Answer: c. randomseed()

Detailed Solution: randomseed() generates the same set of numbers every time a program is run. It depends on the argument passed. (Please refer Lecture 23@9:21)

QUESTION 15:

How many types of interrupt are possible in an Arduino?

a. 1
b. 2
c. 3
d. 4

Correct Answer: b. 2

Detailed Solution: 2 types of interrupt: hardware and software. (Please refer Lecture 23@10:26)

**********END**********

Introduction to

Internet of Things

Assignment-Week 6

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

In Python programming, what is used to indicate the different blocks of code?

a. Curly braces
b. Indentation
c. Square brackets
d. First brackets

Correct Answer: b. Indentation

Detailed Solution: Python programming uses indentation to indicate the different blocks of code. (Please refer Lecture 26@7:34)

QUESTION 2:

How many data types does Python support?

a. 3
b. 4
c. 5
d. 6

Correct Answer: c. 5

Detailed Solution: Python supports 5 data types: Numbers, Strings, List, Tuple, and Dictionary. (Please refer Lecture 26@11:14)

QUESTION 3:

What is the keyword for else if condition in Python?

a. else if
b. else
c. elseif
d. elif

Correct Answer: d. elif

Detailed Solution: Python uses the elif keyword for else if. (Please refer Lecture 26@16:04)

QUESTION 4:

What is the keyword for ending the current iteration of a loop and start a new iteration?

a. break
b. continue
c. exit
d. stop

Correct Answer: b. continue

Detailed Solution: The continue keyword in python is used for ending the current iteration of a loop and start a new iteration. (Please refer Lecture 26@17:44)

QUESTION 5:

How are functions with return values defined in python?

a. void
b. int
c. def
d. float

Correct Answer: c. def

Detailed Solution: With or without return values, python functions are defined with def. (Please refer Lecture 26@18:34)

QUESTION 6:

What are variables that are declared and has its scope within a function in python called?

a. Global variable
b. Local variable
c. Hierarchical variable
d. Dangling variable

Correct Answer: b. Local variable

Detailed Solution: Local variables are defined within a function and its scope is limited to it. (Please refer Lecture 26@25:24)

QUESTION 7:

Which of the following arguments is used for both file read and write operation in python?

a. r+
b. r-
c. r*
d. r#

Correct Answer: a. r+

Detailed Solution: r+ indicates both read and write operations. (Please refer Lecture 27@2:38)

QUESTION 8:

Which of the following communication modules does a Raspberry Pi device support?

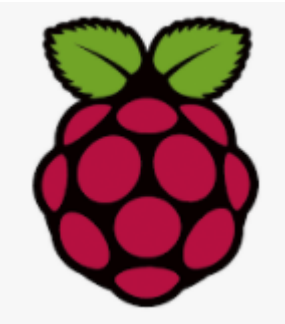
a. Ethernet
b. WiFi
c. Bluetooth
d. All of these

Correct Answer: d. All of the these

Detailed Solution: Raspberry Pi supports Ethernet, WiFi, and Bluetooth. (Please refer Lecture 28@6:22)

QUESTION 9:

What device does this image represent?



a. Arduino
b. Lilypad
c. Raspberry pi
d. NodeMCU

Correct Answer: c. Raspberry pi

Detailed Solution: Raspberry pi

QUESTION 10:

For enabling a raspberry pi camera, which option to select after raspi-config?

a. Boot options
b. Interfacing options
c. Advanced options
d. Localization options

Correct Answer: b. Interfacing options

Detailed Solution: For enabling the camera, choose Interfacing options after raspi-config. (Please refer Lecture 29@18:51)

QUESTION 11:

In Raspberry pi, what is the command for capturing an image?

a. raspicapture
b. raspicam
c. raspisnap
d. raspistill

Correct Answer: d. raspistill

Detailed Solution: raspistill –o imagename is used to capture images in raspberry pi. (Please refer Lecture 29@19:31)

QUESTION 12:

In a DHT22 sensor, which pin number gives NULL?

a. PIN 1
b. PIN 2
c. PIN 3
d. PIN 4

Correct Answer: PIN 3

Detailed Solution: In a DHT sensor, PIN 1 is power supply, PIN 2 is data, PIN 3 is NULL and PIN 4 is GND. (Please refer Lecture 30@5:11)

QUESTION 13:

For a downloaded python library in raw format (like Github for instance), what is the python command to install the package in the raspberry pi terminal?

a. sudo python setup.py install
b. sudo python install setup.py
c. sudo install python setup.py
d. sudo setup.py python install

Correct Answer: a. sudo python setup.py install

Detailed Solution: sudo python setup.py install. (Please refer Lecture 30@8:41)

QUESTION 14:

What is the linux command for listing contents of a folder/directory?

a. dir
b. ls
c. mkdir
d. cd

Correct Answer: b. ls

Detailed Solution: The ls command lists the contents of a folder/directory.

QUESTION 15:

True or False?

The Raspberry Pi device supports only the Raspbian OS.

a. True
b. False

Correct Answer: b. False

Detailed Solution: The Raspberry Pi device supports any arm64 OS.

**********END**********

Introduction to Internet of Things

Assignment-Week 7

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

On what basis is SDN network management simplified?

a. Device
b. Network
c. Traffic
d. Flow

Correct Answer: a. Device

Detailed Solution: SDN network management is simplified based on user, device, and application requirements. (Please refer Lecture 35@2:45)

QUESTION 2:

Which of the following is not a component of policies in SDN?

a. Controller
b. Gateway
c. Orchestration
d. Service logic

Correct Answer: b. Gateway

Detailed Solution: Policies in SDN consist of the SDN controller, orchestration, and service logic (Please refer Lecture 35@4:35)

QUESTION 3:

Which of the following layers in the SDN architecture is responsible for rule placement and traffic engineering at the backbone networks?

a. Constrained node networks
b. Intelligent aggregation

c. Transport Network
d. IoT network

Correct Answer: c. Transport Network

Detailed Solution: The transport network in the SDN architecture is responsible for rule placement and traffic engineering at the backbone networks. (Please refer Lecture 35@6:55)

QUESTION 4:

Fill in the blank:

The Soft-WSN method increases the packet delivery ratio as well as _____.

a. Data packets
b. Control packets
c. Replicated packets
d. Fragmented packets

Correct Answer: b. Control packets

Detailed Solution: Soft-WSN reduces the number of replicated data packets and increases the packet delivery ratio. However, it increases the number of control packets. (Please refer Lecture 35@13:00)

QUESTION 5:

Which of the following programming languages does SDN-WISE use for programming the nodes?

a. C/C++
b. Python
c. Java
d. Any programming language

Correct Answer: d. Any programming language

Detailed Solution: Using an API, the nodes in SDN-WISE may be programmed using any language in real-time. (Please refer Lecture 35@13:55)

QUESTION 6:

In the SDN-WISE protocol stack, which layer contains the flow rules?

a. Forwarding layer
b. Routing layer
c. Topology manager
d. Emulated node

Correct Answer: a. Forwarding layer

Detailed Solution: The forwarding layer in SDN-WISE contains the flow rules. (Please refer Lecture 35@14:25)

QUESTION 7:

Which of the following APIs in SDN uses OpenFlow protocol?

a. Southbound API
b. Northbound API
c. East-Westbound API
d. None of these

Correct Answer: a. Southbound API

Detailed Solution: The southbound API is used to communicate between control and infrastructure layers and uses the OpenFlow protocol. (Please refer Lecture 34@1:41)

QUESTION 8:

Fill in the blank:

Controllers in SDN receive requests from the _____.

a. End users
b. Gateways
c. Switches
d. Routers

Correct Answer: c. Switches

Detailed Solution: SDN controllers should be able to handle all requests from the SDN switches (Please refer Lecture 34@2:41)

QUESTION 9:

How many deployment architectures are possible in SDN?

a. 1
b. 2
c. 3
d. 4

Correct Answer: d. 4

Detailed Solution: SDN supports flat, hierarchical, ring, and mesh architectures. (Please refer Lecture 34@5:06)

QUESTION 10:

Among centralized and distributed systems, which of them may be considered as unstable?

a. Centralized
b. Distributed
c. Neither centralized nor distributed
d. Cannot be determined

Correct Answer: a. Centralized

Detailed Solution: The centralized systems are very unstable, since any problem that affects the central server can generate chaos throughout the system. However, the distributed systems are more stable, by storing the totality of the system information in a large number of nodes that maintain equal conditions with each other.

QUESTION 11:

Which of the following types of controller does Mininet support?

a. Remote controller
b. Local controller
c. Neither remote nor local controller

d. Both remote and local controller

Correct Answer: d. Both remote and local controller

Detailed Solution: Mininet supports both remote and local controllers. (Please refer Lecture 34@12:21)

QUESTION 12:

Which of the following effects the performance of SDN?

a. Rule placement
b. Controller placement
c. Neither rule nor controller placement
d. Both rule and controller placement

Correct Answer: d. Both rule and controller placement

Detailed Solution: Performance of SDN depends on both rule and controller placement. (Please refer Lecture 34@13:06)

QUESTION 13:

SDN helps in separating the data and control planes. However, it also provides provisions to interact with the application layer.

a. True
b. False

Correct Answer: a. True

Detailed Solution: In SDN, interaction between the data, control, and application layers is possible using APIs. (Please refer Lecture 33@14:53)

QUESTION 14:

Under what condition may the SDN network be congested with PACKET-IN messages?

a. Large number of SDN controllers
b. Fewer number of SDN controllers
c. Does not matter

d. SDN network does not get congested

Correct Answer: b. Fewer number of SDN controllers.

Detailed Solution: As the flow rules keep getting replaced, the network will be congested with PACKET-IN messages in case of lesser number of controllers. (Please refer Lecture 34@5:01)

QUESTION 15:

True or False?

The OpenFlow protocol supports Flow Rules in any format.

a. True
b. False

Correct Answer: b. False

Detailed Solution: The OpenFlow protocol supports Flow Rules in only specific formats. (Please refer Lecture 33@18:13)

**********END**********

Introduction to Internet of Things

Assignment-Week 8

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

True or False?

Although SDN and OpenFlow have their own data packet routing methods, they are dependent on core routing policies.

a. True
b. False

Correct Answer: b. False

Detailed Solution: SDN networking is not dependent on core routing policies. (Please refer Lecture 36@3:41)

QUESTION 2:

A signal with power 3 mW is distorted by noise with power 0.001 mW and interference of 0.003 mW. What is the signal to interference plus noise ratio (SINR) in dB?

a. 38.75
b. 3.875
c. 28.75
d. 2.875

Correct Answer: c. 28.75

Detailed Solution: SINR in dB = 10 * $\log_{10}$ (S/I+N) = 28.75 dB and 20 * $\log_{10}$ (S/I+N) = 57.5 Watts (Please refer Lecture 36@6:12)

QUESTION 3:

In ODIN, which of the following components are placed on the physical devices?

a. Odin agent
b. Odin master
c. Either Odin agent or Odin master
d. Neither Odin agent nor Odin master

Correct Answer: a. Odin agent

Detailed Solution: Odin agent is placed with the physical devices. (Please refer Lecture 36@9:53)

QUESTION 4:

In ODIN, which of the following is responsible for assigning light virtual access points to the clients?

a. Odin agent
b. Odin master
c. Either Odin agent or Odin master
d. Neither Odin agent nor Odin master

Correct Answer: b. Odin master

Detailed Solution: Odin master is responsible for assigning light virtual access points to the physical devices. (Please refer Lecture 36@10:33)

QUESTION 5:

Which of the following terms refers to the network decomposition into relatively independent subnets for independent operations and optimization?

a. Network partitioning
b. Network matching
c. Load balancing
d. Fault tolerance

Correct Answer: a. Network partitioning

Detailed Solution: Network partitioning refers to the network decomposition into relatively independent subnets for independent operations and optimization. (Please refer Lecture 36@11:13)

QUESTION 6:

Compared to conventional methods, which of the following does the Mobi-Flow method help in reducing?

a. Message overhead
b. Energy consumption
c. Both message overhead and energy consumption
d. Neither message overhead nor energy consumption

Correct Answer: c. Both message overhead and energy consumption

Detailed Solution: Mobi-Flow reduces both message overhead and energy consumption. (Please refer Lecture 36@16:43)

QUESTION 7:

In data center networking, when should the flows be classified for adequately forwarding them on the network?

a. Before inserting the flow rules
b. On inserting the flow rules
c. After inserting the flow rules
d. None of these

Correct Answer: a. Before inserting the flow rules

Detailed Solution: The the flows should be classified before inserting the flow rules in the switches for adequately forwarding them on the network. (Please refer Lecture 36@17:43)

QUESTION 8:

Which of the following consists of homogeneous computing nodes?

a. Cloud computing
b. Utility computing
c. Grid computing
d. Cluster computing

Correct Answer: d. Cluster computing

Detailed Solution: Cluster computing consists of homogeneous computing nodes. (Please

refer Lecture 37@5:01)

QUESTION 9:

According to NIST, how many essential characteristics exist in cloud computing model?

a. 3
b. 4
c. 5
d. 6

Correct Answer: c. 5

Detailed Solution: Cloud computing has 5 essential characteristics. (Please refer Lecture 37@13:31)

QUESTION 10:

Which of the following features in cloud computing refers to rapid and automatic resource allocation?

a. Resource pooling
b. Measured service
c. On-demand service
d. Rapid Elasticity

Correct Answer: d. Rapid elasticity

Detailed Solution: Rapid elasticity in cloud computing refers to rapid and automatic resource allocation. (Please refer Lecture 37@20:31)

QUESTION 11:

In cloud computing, which of the following components contains SaaS?

a. Application
b. Platform
c. Storage
d. Infrastructure

Correct Answer: a. Application

Detailed Solution: The Applications component contains Web Apps, SaaS, etc. (Please refer Lecture 37@23:32)

QUESTION 12:

Which of the following helps in executing consumer-created or acquired applications onto cloud infrastructure?

a. SaaS
b. PaaS
c. IaaS
d. None of these

Correct Answer: b. PaaS

Detailed Solution: PaaS helps in executing consumer-created or acquired applications onto cloud infrastructure. (Please refer Lecture 37@26:22)

QUESTION 13:

True or False?

Public cloud provides more security than private cloud.

a. True
b. False

Correct Answer: b. False

Detailed Solution: Private cloud provides more security than public cloud. (Please refer Lecture 37@31:22)

QUESTION 14:

Which of the following cloud computing models does AWS EC2 belong to?

a. IaaS
b. PaaS
c. SaaS

d. None of these

Correct Answer: a. IaaS.

Detailed Solution: AWS EC2 is a popular example of IaaS.

QUESTION 15:

Which of the following must include both private and public clouds?

a. Multicloud
b. Hybrid cloud
c. Public cloud
d. All of the above

Correct Answer: b. Hybrid cloud

Detailed Solution: Multicloud may or may not involve private clouds. However, Hybrid cloud must have both private and public cloud models. (Please refer Lecture 37@32:02)

**********END**********

Introduction to Internet of Things

Assignment-Week 9

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

How many components does the OpenStack Horizon dashboard consist of?

a. 7
b. 8
c. 9
d. 10

Correct Answer: b. 8

Detailed Solution: The OpenStack Horizon dashboard consists of 8 components. (Please refer Lecture 41@3:55)

QUESTION 2:

Which of the following components is responsible for providing identity service in OpenStack?

a. Nova
b. Ceilometer
c. Keystone
d. Neutron

Correct Answer: c. Keystone

Detailed Solution: Keystone is responsible for providing identity service in OpenStack. (Please refer Lecture 41@4:00)

QUESTION 3:

What is the function of Neutron in OpenStack?

a. Identity service
b. Object storage
c. Networking service
d. Billing

Correct Answer: c. Networking service

Detailed Solution: Neutron in OpenStack is responsible for providing networking service. (Please refer Lecture 41@5:20)

QUESTION 4:

For uninstalling OpenStack, which file in the devstack directory should be executed?

a. stack.sh
b. unstack.sh
c. uninstall_stack.sh
d. detach.sh

Correct Answer: b. unstack.sh

Detailed Solution: For uninstalling OpenStack, unstack.sh in the devstack directory should be executed. (Please refer Lecture 41@10:20)

QUESTION 5:

What is the command (Linux) for generating a public/private authentication key pair?

a. keygen
b. ssh-key
c. ssh-genkey
d. ssh-keygen

Correct Answer: d. ssh-keygen

Detailed Solution: ssh-keygen generates a public/private authentication key pair.

QUESTION 6:

Which of the following methods are used for encrypting the communications in Secure Shell Protocol (SSH)?

a. Symmetric

b. Asymmetric

c. Hashing

d. All of the above

Correct Answer: d. All of the above

Detailed Solution: SSH uses public/asymmetric key cryptography.

QUESTION 7:

In a sensing environment, when do the sensor nodes consume unnecessary energy?

a. Rapidly changing environment
b. Slow changing environment
c. Change in environment does not matter
d. None of the above

Correct Answer: b. Slow changing environment

Detailed Solution: Slow changing environment does not lead to much change in the sensor data and hence is considered as unnecessary energy consumption. (Please refer Lecture 43@20:54)

QUESTION 8:

Which of the following caches are responsible for handing requests from the application layer?

a. Internal cache
b. External cache
c. Either internal or external cache
d. Neither internal nor external cache

Correct Answer: a. Internal cache

Detailed Solution: Internal cache is responsible for handling user/application layer requests. (Please refer Lecture 43@21:04)

QUESTION 9:

What is responsible for dealing with the usage of physical sensor nodes for pricing in sensor cloud?

a. pH
b. pI
c. pS
d. pHI

Correct Answer: a. pH

Detailed Solution: Pricing attributed to hardware (pH) is responsible for dealing with the usage of physical sensor nodes. (Please refer Lecture 43@30:05)

QUESTION 10:

In a group of fog nodes, if one is selected at random for task offloading, the probability that it accepts the task is 0.6 and the probability that it accepts and completes the task is 0.3. If a fog node that accepts the task is selected at random, what is the probability that it completes the execution?

a. 0.2
b. 0.3
c. 0.4
d. 0.5

Correct Answer: d. 0.5

Detailed Solution: A: Accepts task, C: Completes task

P(A) = 0.6, P(A and C) = 0.3

P(A|C) = P(A and C)/P(C) = 0.3/0.6 = 0.5

QUESTION 11:

True or False?

The fog computing platform is not a replacement for the cloud. It is only used to enhance the performance of the cloud.

a. True
b. False

Correct Answer: a. True

Detailed Solution: The fog computing platform is not a replacement for the cloud. It is only used to enhance the performance of the cloud. (Please refer Lecture 45@1:51)

QUESTION 12:

Typically, what type of storage is generally offered by fog nodes?

a. Transient
b. Permanent
c. Historical analysis
d. All of the above

Correct Answer: a. Transient

Detailed Solution: Typically, fog nodes provide transient storage. (Please refer Lecture 45@9:38)

QUESTION 13:

In the fog computing architecture, which of the following devices sends a summary to the cloud for historical analysis and storage?

a. End user devices
b. Fog node
c. Aggregate fog node
d. All of these

Correct Answer: d. All of these

Detailed Solution: The end user devices, fog nodes, and aggregator fog nodes send summary to the cloud for historical analysis and storage. (Please refer Lecture 45@13:28)

QUESTION 14:

Choose the correct option for geographical coverage.

a. Fog node: Very local, Fog aggregator node: Global, Cloud: Wider
b. Fog node: Wider, Fog aggregator node: Very local, Cloud: Global
c. Fog node: Very local, Fog aggregator node: Wider, Cloud: Global

d. Fog node: Global, Fog aggregator node: Very local, Cloud: Wider

Correct Answer: c. Fog node: Very local, Fog aggregator node: Wider, Cloud: Global

Detailed Solution: Fog node: Very local, Fog aggregator node: Wider, Cloud: Global. (Please refer Lecture 45@17:28)

QUESTION 15:

Which of the following is true for fog nodes?

a. Mobile
b. Static
c. May be mobile or static
d. None of the above

Correct Answer: c. May be mobile or static

Detailed Solution: Fog nodes may be both mobile or static. (Please refer Lecture 45@25:28)

**********END**********

Introduction to Internet of Things

Assignment-Week 10

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

What do smart cities use for increasing the availability of their infrastructure and services to the public?

a. Information technology
b. Communication technology
c. Information and communication technology
d. None of the above

Correct Answer: c. Information and communication technology

Detailed Solution: Smart cities use information and communication technology to increase the availability of their infrastructure and services to the public. (Please refer Lecture 46@8:42)

QUESTION 2:

In the context of humans, which among the following is analogous to software in smart cities?

a. Brain
b. Cognition
c. Sensory organs
d. Nerves

Correct Answer: b. Cognition

Detailed Solution: Cognition in humans is analogous to software in smart cities. (Please refer Lecture 46@8:52)

QUESTION 3:

In smart cities, which application area focuses on Innovative spirit, entrepreneurship, economic image & trademarks, productivity, flexibility or labour market, international embeddedness and ability to transform?

a. Smart Economy
b. Smart Governance
c. Smart People
d. Smart Environment

Correct Answer: a. Smart Economy

Detailed Solution: Smart Economy focuses on Innovative spirit, entrepreneurship, economic image & trademarks, productivity, flexibility or labour market, international embeddedness and ability to transform. (Please refer Lecture 46@11:02)

QUESTION 4:

Cities, Localities, and Population are components of which of the following components in smart cities?

a. Smart Governance
b. Smart People
c. Smart Living
d. Smart Mobility

Correct Answer: d. Smart Mobility

Detailed Solution: Cities, Localities, and Population are components of Smart Mobility in Smart Cities. (Please refer Lecture 46@18:32)

QUESTION 5:

Which of the following is not a data fusion stage?

a. Decision level
b. Feature level
c. Signal level
d. None of these

Correct Answer: d. None of these

Detailed Solution: Data fusion stages are: Decision, Feature, Pixel, and Signal levels
See lecture 47 @ 10:53

QUESTION 6:

Which IoT challenge in smart cities covers device failures?

a. Reliability
b. Large scale
c. Heterogeneity
d. Security and privacy

Correct Answer: a. Reliability

Detailed Solution: The reliability challenge covers device failures in smart cities, (Please refer Lecture 46@32:42)

QUESTION 7:

What is the process of combining data from several sources to produce new data called?

a. Data collection
b. Data fusion
c. Information fusion
d. Data synthesis

Correct Answer: b. Data fusion

Detailed Solution: Data fusion is the process of combining data from several sources to produce new data. (Please refer Lecture 47@5:27)

QUESTION 8:

Which of the following challenges covers outliers and missing data?

a. Imperfection
b. Conflicts
c. Ambiguity
d. Alignment

Correct Answer: c. Ambiguity

Detailed Solution: Ambiguity covers outliers and missing data. (Please refer Lecture 47@8:07)

QUESTION 9:

Typically, in a smart vehicle, GPS connects to which one of the following?

a. Router
b. Gateway
c. Any WiFi device
d. Satellite

Correct Answer: d. Satellite

Detailed Solution: Typically, GPS connect to the satellites. (Please refer Lecture 47@13:27)

QUESTION 10:

True or False?
Short parking search time leads to customer savings.

a. True
b. False

Correct Answer: b. False

Detailed Solution: Short parking search time leads to more parked time, leading to increase

in revenue for the service provider. (Please refer Lecture 47@15:27)

QUESTION 11:

Which of the following is true for smart vehicles?

a. They can communicate with other devices within the vehicle
b. They can communicate with other smart vehicles
c. They can communicate with nearby fixed infrastructures
d. All of these

Correct Answer: d. All of these

Detailed Solution: Smart vehicles can communicate with devices within the vehicle, other smart vehicles, and nearby fixed infrastructures. (Please refer Lecture 49@4:36)

QUESTION 12:

Typically, what is the nature of a V2X architecture?

a. Centralized
b. Decentralized
c. Distributed
d. Does not matter

Correct Answer: c. Distributed

Detailed Solution: V2X deployments follow a distributed architecture. (Please refer Lecture 49@11:26)

QUESTION 13:

True or false?
VANETS derive their routing protocols from MANETS.

a. True
b. False

Correct Answer: a. True

Detailed Solution: VANETS derive their routing protocols from MANETS. (Please refer Lecture 49@16:26)

QUESTION 14:

Which of the following best describes the duration of the network links in a vehicular network?

a. Long
b. Short
c. Slow
d. Medium

Correct Answer: b. Short

Detailed Solution: In vehicular networks, the network links are alive for a short duration due to high mobility (Please refer Lecture 49@17:26)

QUESTION 15:

What is the primary component in content centric networking?

a. Name
b. Location
c. IP address
d. Mobility

Correct Answer: a. Name

Detailed Solution: Name is a primary component of content centric networking. (Please refer Lecture 49@20:16)

**********END**********

Introduction to Internet of Things

Assignment-Week 11

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

What is a cyber-physical system equipped with sustainable models of energy production, distribution, and usage?

a. Information and communication technology
b. Grid
c. Smart grid
d. Traditional grid

Correct Answer: c. Smart grid

Detailed Solution: Smart grid is a cyber-physical system equipped with sustainable models of energy production, distribution, and usage. (Please refer Lecture 51@1:12)

QUESTION 2:

In traditional electrical grids, which of the following is done manually?

a. Monitoring
b. Restoration
c. Both monitoring and restoration
d. Neither monitoring nor restoration

Correct Answer: c. Both monitoring and restoration

Detailed Solution: In traditional electrical grids, both monitoring and restoration are done manually. (Please refer Lecture 51@1:12)

QUESTION 3:

Which of the following is true for communications in smart grids?

a. One-way communication
b. Two-way communication
c. Three-way communication
d. Four-way communication

Correct Answer: b. Two-way communication

Detailed Solution: Smart grids follow a two-way communication infrastructure. (Please refer Lecture 51@7:22)

QUESTION 4:

True or False?

Smart grids help the customers in reducing their power/electricity costs.

a. True
b. False

Correct Answer: a. True

Detailed Solution: Smart grids help the customers in reducing their power/electricity costs. (Please refer Lecture 51@10:57)

QUESTION 5:

Typically, smart grid benefits which of the following participants?

a. Consumers
b. Service providers
c. Stakeholders
d. All of these

Correct Answer: d. All of these

Detailed Solution: Smart grids benefit both consumers and service providers/stakeholders. (Please refer to the paper in Lecture 51@10:57)

QUESTION 6:

What time is best suitable for running smart devices in order to reduce electricity bills?

a. Peak hours
b. On-peak hours
c. Off-peak hours
d. Does not matter

Correct Answer: c. Off-peak hours

Detailed Solution: Running smart devices in off-peak hours helps in reducing electricity bills, (Please refer Lecture 51@14:13)

QUESTION 7:

True or False?

Smart grids offer homogeneous pricing options to the customers.

a. True
b. False

Correct Answer: b. False

Detailed Solution: Smart grids offer homogeneous pricing options. (Please refer Lecture 51@14:13)

QUESTION 8:

Which of the following properties in smart grids covers load forecasting?

a. Power system efficiency
b. Distributed generation
c. Power quality
d. Consumer participation

Correct Answer: c. Power quality

Detailed Solution: The power quality property of smart grids covers load forecasting. (Please refer Lecture 51@21:38)

QUESTION 9:

Which of the following best represents the relation between IoT, IIoT, and Industry 4.0?

a. IoT = IIoT = Industry 4.0
b. IIoT = IoT + Industry 4.0
c. IIoT = IoT ∩ Industry 4.0
d. IIoT = IoT ∪ Industry 4.0

Correct Answer: c. IIoT = IoT ∩ Industry 4.0

Detailed Solution: IIoT is the intersection of IoT and Industry 4.0. (Please refer Lecture 53@3:51)

QUESTION 10:

Which of the following best represents Industrial Internet of Things?

a. Consumer IoT
b. Enterprise IoT
c. Industry 4.0
d. None of the above

Correct Answer: b. Enterprise IoT

Detailed Solution: IIoT may be referred as Enterprise IoT and general IoT as Consumer IoT. (Please refer Lecture 53@5:21)

QUESTION 11:

True or False?

IIoT includes Machine-to-Machine communications.

a. True
b. False

Correct Answer: a. True

Detailed Solution: IIoT includes Machine-to-Machine communications. (Please refer Lecture 53@6:31)

QUESTION 12:

Fill in the blank:

IIoT is based on _____ approach.

a. Wrap and re-use
b. Rip and replace
c. Wrap and replace
d. Rap and re-use

Correct Answer: a. Wrap and re-use

Detailed Solution: IIoT is based on wrap and re-use approach than rip and replace. (Please refer Lecture 53@9:31)

QUESTION 13:

Which of the following may be used to describe IIoT?

a. Mechanized production
b. Mass production
c. Internet evolution and automation
d. None of these

Correct Answer: d. None of these

Detailed Solution: IIoT is the fourth industrial revolution. (Please refer Lecture 53@10:21)

QUESTION 14:

Typically, how many types of data exist?

a. 1
b. 2
c. 3
d. 4

Correct Answer: b. 2

Detailed Solution: Data may be structured or unstructured. (Please refer Lecture 55@9:51)

QUESTION 15:

Which of the following V's in big data refers to the quantity of data that is generated?

a. Volume
b. Velocity
c. Veracity
d. Value

Correct Answer: a. Volume

Detailed Solution: Volume refers to the quantity of data that is generated. (Please refer Lecture 55@10:51)

**********END**********

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 12

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

What is the process of examining data sets in order to draw conclusions called?

a. Data fusion
b. Data fission
c. Data analysis
d. Data analytics

Correct Answer: d. Data analytics

Detailed Solution: Data analytics is the process of examining data sets in order to draw conclusions. (Please refer Lecture 56@2:25)

QUESTION 2:

How many types of data analysis exist?

a. 1
b. 2
c. 3
d. 4

Correct Answer: b. 2

Detailed Solution: Data analysis are of 2 types: Qualitative and Quantitative. (Please refer Lecture 56@3:15)

QUESTION 3:

Notice, collect, and think about things are principles of which category of data analysis?

a. Qualitative
b. Quantitative
c. Both qualitative and quantitative
d. Neither qualitative nor quantitative

Correct Answer: a. Qualitative

Detailed Solution: Notice, collect, and think about things are principles of qualitative analysis. (Please refer Lecture 56@3:45)

QUESTION 4:

Analysis such as mean, median, mode, and others are methods used in which type of data analysis?

a. Qualitative
b. Quantitative
c. Both qualitative and quantitative
d. Neither qualitative nor quantitative

Correct Answer: b. Quantitative

Detailed Solution: Analysis such as mean, median, mode, and others are methods of quantitative analysis. (Please refer Lecture 56@4:35)

QUESTION 5:

Which method of data analysis involves observing the data?

a. Qualitative
b. Quantitative
c. Both qualitative and quantitative
d. Neither qualitative nor quantitative

Correct Answer: a. Qualitative

Detailed Solution: Qualitative analysis involves observing the data. (Please refer Lecture 56@5:35)

QUESTION 6:

Choose the right option for the ordered pair (P, X) in a statistical model.

a. P: set of all possible observations, X: set of probability distributions
b. P: set of all probability distributions, X: set of possible observations
c. P: set of all probability values, X: set of possible distributions
d. P: set of possible distributions, X: set of all probability values

Correct Answer: a. P: set of all possible observations, X: set of probability distributions

Detailed Solution: For the ordered pair (P,X) in a statistical model, P: set of all possible observations, X: set of probability distributions (Please refer Lecture 56@7:25)

QUESTION 7:

Which of the following is true for a complete model with V number of variables and E number of equations?

a. $V < E$
b. $V > E$
c. $V = E$
d. $V \neq E$

Correct Answer: c. V = E

Detailed Solution: A complete model has the same number of equations as the number of variables. (Please refer Lecture 56@8:15)

QUESTION 8:

How many categorical factors are necessary while performing an ANOVA?

a. 1
b. 2
c. Atleast 1
d. Atleast 2

Correct Answer: c. Atleast 1

Detailed Solution: ANOVA must have a continuous response variable and atleast one categorical factor with two or more levels. (Please refer Lecture 56@8:55)

QUESTION 9:

True or False?

The variance between the groups in ANOVA should be the same.

a. True
b. False

Correct Answer: a. True

Detailed Solution: The variance between the groups in ANOVA should be the same. (Please refer Lecture 56@11:06)

QUESTION 10:

What is the value of F-ratio when the null hypothesis is false?

a. 0
b. 1
c. < 1
d. > 1

Answer: d. > 1

Detailed Answer: The F-ratio is greater than 1 when the null hypothesis is false. Please refer Lecture 56@12:39

QUESTION 11:

In AgriSens, which of the following is responsible for sending field information to a farmer's cell using SMS?

a. Repository data server
b. Web server
c. Multi users server
d. None of the above

Correct Answer: c. Multi users server

Detailed Solution: The multi users servers are responsible for sending text messages and answering farmer queries. (Please refer Lecture 57@9:25)

QUESTION 12:

In the healthcare use case, what does lower error imply?

a. Increased efficiency, reduced cost
b. Reduced efficiency, increased cost
c. Equal efficiency and cost
d. None of the above

Correct Answer: a. Increased efficiency, reduced cost

Detailed Solution: Lower error implies increased efficiency and reduced cost. (Please refer Lecture 58@4:23)

QUESTION 13:

True or False?
Although IoT in healthcare has its advantages, it increases the cost.

a. True
b. False

Correct Answer: b. False

Detailed Solution: IoT in healthcare helps in lowering the cost. (Please refer Lecture 58@12:13)

QUESTION 14:

Typically, how many components exist in IoT healthcare?

a. 1
b. 2
c. 3
d. 4

Correct Answer: d. 4
Detailed Solution: IoT healthcare consists of sensing, aggregated, processing, and cloud platform layers. (Please refer Lecture 58@7:13)

QUESTION 15:

How many axes data does a typical accelerometer in a smartphone provide?

a. 1
b. 2
c. 3
d. 4

Correct Answer: c. 3

Detailed Solution: A typical accelerometer gives 3 axes data.

**********END**********